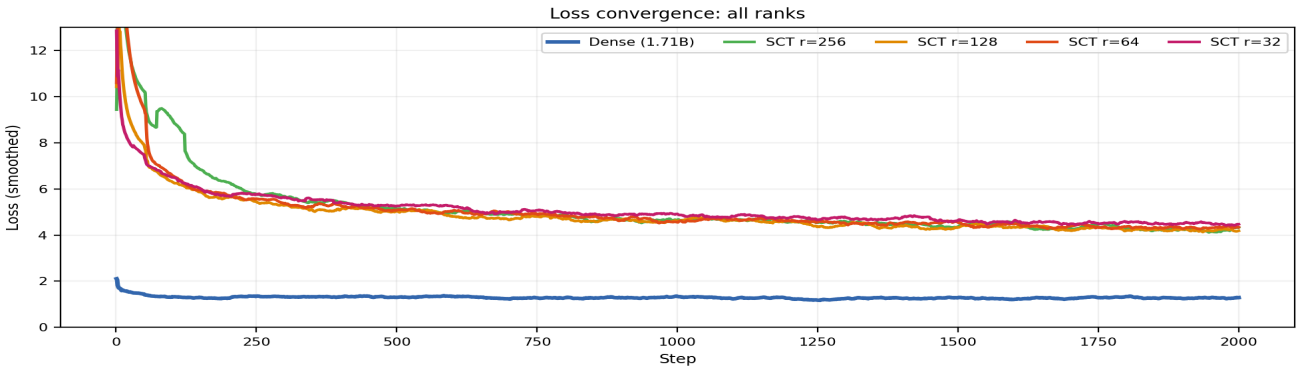
Patent pending PTIE20260000000219 | SmolLM2-1.7B | Rank sweep

SCT Rank Sweep: Compression vs Convergence

Ranks 32, 64, 128, 256 vs dense baseline on Alpaca, 2000 steps, A100 40GB

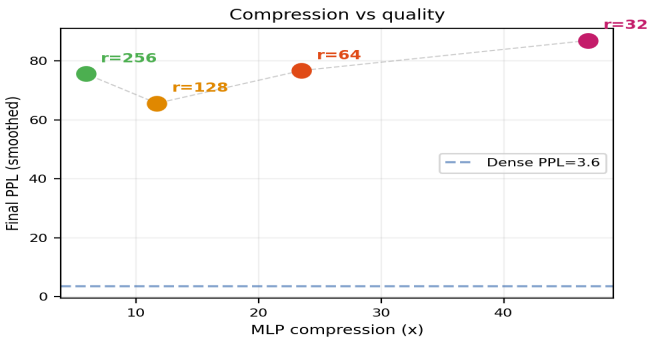
W = U diag(s) V^T with Stiefel QR retraction | github.com/EctoSpace/SCT

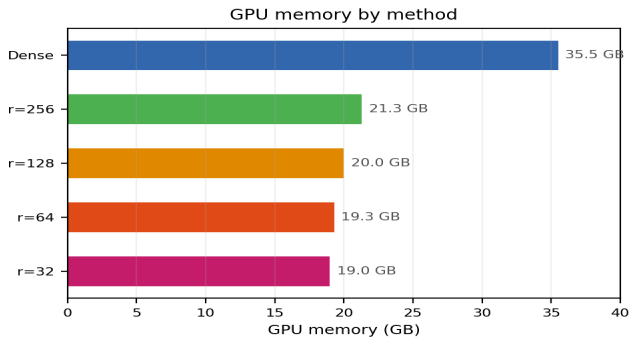


RESULTS SUMMARY

Method	Params	Compression	Loss	PPL	GPU	Step
Dense	1,711M	1.0x	1.29	3.6	35.5 GB	1.17s
SCT r=256	692M	5.9x	4.33	75.6	21.3 GB	1.05s
SCT r=128 *	598M	11.7x	4.18	**65.6**	20.0 GB	0.74s
SCT r=64	551M	23.5x	4.34	76.7	19.3 GB	0.62s
SCT r=32	527M	46.9x	4.47	86.9	19.0 GB	0.56s

* Best SCT result





KEY FINDINGS

All ranks converge to the same loss floor (~4.2-4.5).
Rank 256 (5.9x) and rank 32 (46.9x) end within 0.3 loss of each other. MLP rank is not the bottleneck.

Rank 128 is the sweet spot.
Best PPL (65.6) at 11.7x MLP compression, 20 GB VRAM, 0.74s/step. Rank 256 is worse due to LR overshoot.

The ~3 loss gap points to attention layers, not MLP rank.
Attention (403M params, 77% of model) trained at 25x dense LR. Per-component LR schedule is the next step.

Memory efficiency scales with compression.
GPU drops from 35.5 GB (dense) to 19.0 GB (rank 32), a 46% reduction. Even rank 256 saves 40% VRAM.